

Deterministic geometric Savitzky-Golay dynamics estimator

Shared parameter update: reference -> estimated

mass [kg]	2.351597591	->	2.351597591	delta=+4.42273e-10	ratio=1
CoG x [m]	-0.002024708562	->	-0.002024708584	delta=-2.18959e-11	
ratio=-					
CoG y [m]	-3.052657894e-05	->	-3.052654895e-05	delta=+2.99884e-11	
ratio=-					
CoG z [m]	0.009509749599	->	0.009509749615	delta=+1.6046e-11	
ratio=-					
rotor effectiveness 1	1	->	0.9999999992	delta=-8.1715e-10	ratio=1
rotor effectiveness 2	1	->	0.9999999995	delta=-5.11908e-10	ratio=1
rotor effectiveness 3	1	->	0.9999999991	delta=-9.48705e-10	ratio=1
rotor effectiveness 4	1	->	0.9999999988	delta=-1.17464e-09	ratio=1
command lag [s]	0	->	0.004000000364	delta=+0.004	ratio=-

Inertia matrix [kg m^2]

reference:

[0.06500006148, -7.27899253e-07, 1.901508003e-05]
[-7.27899253e-07, 0.06495265634, 5.9167305e-08]
[1.901508003e-05, 5.9167305e-08, 0.1289921107]

estimated:

[0.06500006149, -7.278992141e-07, 1.901508008e-05]
[-7.278992141e-07, 0.06495265636, 5.916802101e-08]
[1.901508008e-05, 5.916802101e-08, 0.1289921107]

Principal moments: reference -> estimated

principal moment 1 [kg m^2]	0.06495264516	->	0.06495264518	delta=+1.72229e-11	
ratio=1					
principal moment 2 [kg m^2]	0.06500006701	->	0.06500006702	delta=+9.19864e-12	
ratio=1					
principal moment 3 [kg m^2]	0.1289921163	->	0.1289921163	delta=+8.87801e-12	ratio=1

Optimization

joint dynamics loss: 2.73431150253

Gaussian-prior cost: 2.16993511725e-16

selected strict-ZOH lag [s]: 0.004000000364

Bag failure_1_single

reconstruction position RMSE [m]: 0.00156560090701

reconstruction position component RMSE [m]: [0.000908 0.000474 0.001184]

reconstruction orientation RMSE [deg]: 0.845162170593

gyro consistency RMSE xyz [rad/s]: [0.051104 0.102229 0.030994]

specific-force consistency RMSE xyz [m/s^2]: [0.935582 0.147749 0.268613]

inferred external force RMS xyz [N]: [1.829624 0.546727 3.981926]

inferred external torque RMS xyz [N m]: [0.124851 0.557521 0.104406]

bag dynamics loss: 2.73431150253

Recorded command timestamp intervals (data-derived)

Bag failure_1_single

rotor: median dt=0.00499773026 s (200.091 Hz), min=2.00271606e-05 s, max=0.0138893127 s

gimbal: median dt=0.00499868393 s (200.053 Hz), min=4.529953e-06 s, max=2.48282003 s